

Zaibei Li

Human-Centered Computing
Department of Computer Science
University of Copenhagen

zali@di.ku.dk
+45 91858214
lizaibeim.github.io

RESEARCH PROFILE

Doctoral researcher in computer science working at the intersection of multimodal sensing, wearable systems, and human-centered computing. Experienced in designing and building end-to-end systems for multimodal data collection, sensor fusion, interpretable behavioral modeling, and interactive AI applications.

EDUCATION

Ph.D. Computer Science, University of Copenhagen, Copenhagen, 2027 (expected)
M.S. Computer Science, University of Copenhagen, 2022
B.S. Information Technology, Hong Kong Polytechnic University, 2019
Exchange Computer Science, The Korea Advanced Institute of Science and Technology, 2017

RESEARCH AREAS

Multimodal learning analytics: multimodal data collection, behavioral modeling, and collaborative learning
Ubiquitous computing: IoT systems, embedded sensing, and mobile computing
Human-centered computing: interpretable AI, interactive systems, and human-AI interaction

PUBLICATIONS

Manuscripts in Revision

2026 **Z. Li** et al. “CoLA: An Evidence-Continuity System for Multimodal Observation and Support in Collaborative Learning.” *To be resubmitted to Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), November 2026 round [In revision]*

Conference Proceedings

2026 **Z. Li**, V. Holm-Janass, S. Yamaguchi, and D. Spikol. “Designing for Transparency: Gaze-Augmented Collaborative Action Recognition with Vision-Language Models.” *Proceedings of the 26th IEEE International Conference on Advanced Learning Technologies (ICALT ’26)* [Accepted]
2025 **Z. Li**, S. Yamaguchi, and D. Spikol. “OpenMMLA: an IoT-based Multimodal Data Collection Toolkit for Learning Analytics.” *Proceedings of the 15th Learning Analytics and Knowledge Conference (LAK ’25)*, <https://doi.org/10.1145/3706468.3706525>
2024 **Z. Li**, M. T. Jensen, A. Nolte, and D. Spikol. “Field report for Platform mBox: Designing an Open MMLA Platform.” *Proceedings of the 14th Learning Analytics and Knowledge Conference (LAK ’24)*, <https://doi.org/10.1145/3636555.3636872>
2024 A. Ohsaki, D. Spikol, and **Z. Li**. “Combination of Computer and Human Analytics for Multimodal Learning Analysis in a Hackathon.” pp. 127–129 in *Companion Proceedings 14th International Conference on Learning Analytics and Knowledge (LAK ’24)*, LAK ’24 Companion Proceedings

- 2024 Z. Li, D. Spikol, and L. Nøhr. “mBox-audio: Unveiling Conversational Dynamics through Real-Time and Post-Time Audio Analysis for MMLA.” *pp. 130–132 in Companion Proceedings 14th International Conference on Learning Analytics and Knowledge (LAK ’24)*, LAK ’24 Companion Proceedings
- 2024 D. Spikol, Z. Li, A. Nolte, A. Ohsaki, and K. Rapur. “Investigating Hackathons with Collaboration Analytics.” *Proceedings of the 8th International Conference on Game Jams, Hackathons and Game Creation Events (ICGJ ’24)*, <https://doi.org/10.1145/3697789.3697797>
- 2024 H. Ouhaichi, D. Spikol, Z. Li, and B. Vogel. “Conceptual Design of Multimodal Learning Analytics for Spoken Language Acquisition.” *Proceedings of the 19th European Conference on Technology Enhanced Learning (EC-TEL ’24)*, https://doi.org/10.1007/978-3-031-72312-4_19
- 2024 V. Holm-Janass, O. Caro Miya Marshall, Z. Li, J. Bruun, and D. Spikol. “Design Framework for Multimodal Learning Analytics Leveraging Human Observations.” *Proceedings of the 19th European Conference on Technology Enhanced Learning (EC-TEL ’24)*, https://doi.org/10.1007/978-3-031-72312-4_13
- 2024 H. Ouhaichi, D. Spikol, B. Vogel, and Z. Li. “Analytics in Glocal Classrooms: Integrating Multimodal Learning Analytics in a Smart Learning Environment.” *2024 IEEE International Conference on Advanced Learning Technologies (ICALT ’24)*, <https://doi.org/10.1109/ICALT61570.2024.00033>
- 2024 M. Misfeldt, L. Nøhr, I. Laug, V. Holm-Janass, Z. Li, A. Lorelei Traxler, J. Bruun, D. Spikol, and R. Müller Kristensen. “Addressing Shared Reflective Persistence Through Three Different Lenses.” *pp. 199–202 in supplementary proceedings of the Sixth International Conference on Quantitative Ethnography (ICQE ’24)*, ICQE ’24 Supplementary Proceedings

Reports and Other Publications

- 2026 Z. Li, Q. Li, S. Yamaguchi, and D. Spikol. “BadgeX: IoT-Enhanced Wearable Analytics Meets LLMs for Collaborative Learning.” *arXiv preprint*, arXiv:2604.04093
- 2022 D. Spikol, Z. Li, S. Serrano-Iglesias, H. Ouhaichi, and B. Vogel. “MBOX Lightweight Voice Analysis Sensors for MMLA.” *CROSSMMLA & SLE Workshop: Learning Analytics for Smart Learning Environments Crossing Physical and Virtual Learning Spaces, LAK ’22, Online, Mar 2022*, CrossMMLA @ LAK ’22
- 2022 S. Serrano-Iglesias, D. Spikol, M. Bote-Lorenzo, E. Gómez-Sánchez, and Z. Li. “Challenges of the Adoption of MultiModal Learning Analytics in Smart Learning Environments.” *CROSSMMLA & SLE Workshop: Learning Analytics for Smart Learning Environments Crossing Physical and Virtual Learning Spaces, LAK ’22, Online, Mar 2022*, CrossMMLA @ LAK ’22

CONFERENCE ACTIVITY

Conference Presentations¹

- 2026 Z. Li, V. Holm-Janass, S. Yamaguchi, and D. Spikol. “Designing for Transparency: Gaze-Augmented Collaborative Action Recognition with Vision-Language Models.” Presented at the 26th IEEE International Conference on Advanced Learning Technologies (ICALT ’26), Hung Yen, Vietnam, Jul 6–9.
- 2025 Z. Li, S. Yamaguchi, and D. Spikol. “OpenMMLA: an IoT-based Multimodal Data Collection Toolkit for Learning Analytics.” Presented at the 15th Learning Analytics and Knowledge Conference (LAK ’25), Dublin, Ireland, Mar 3–7.
- 2024 Z. Li, M. T. Jensen, A. Nolte, and D. Spikol. “Field report for Platform mBox: Designing an Open MMLA Platform.” Presented at the 14th Learning Analytics and Knowledge Conference (LAK ’24), Kyoto, Japan, Mar 18–22.

¹Presenting author *italicized* if other than first author.

Session Chair

2026 Chaired Session 2-8B (Big Data in Education and Learning Analytics #2) at the 26th IEEE International Conference on Advanced Learning Technologies (ICALT '26), Hung Yen, Vietnam, Jul 7.

GRANTS AND AWARDS

Awards and Honors

2026 Meta Project Aria Research Partnership / Hardware Grant
2026 Best Full Paper Nomination @ ICALT '26
2025 Best Short Paper Award @ LAK '25
2017 Wong Tit-shing Student Exchange Scholarship

Competitive Grants and Fellowships

2024–2027 PhD Fellowship, University of Copenhagen

RESEARCH EXPERIENCE

Hiroshima City University
Visiting Researcher

Hiroshima, Japan
Nov 2025–Jan 2026

- Prototyped an egocentric multimodal sensing setup using mobile devices, embedded sensors, and smart glasses for real-world collaborative learning analysis.
- Developed CoLA, a wearable platform with an interactive AI facilitator for real-time collaborative learning analytics; established a research collaboration with Meta Project Aria.

University of Copenhagen
Doctoral Researcher

Copenhagen, Denmark
Feb 2024–Present

- Designed and developed OpenMMLA, a reusable IoT-based toolkit for multimodal data collection, synchronization, and analytics in collaborative learning environments.
- Built interactive pipeline components for sensor fusion, behavioral modeling, and visualization, enabling interpretable analysis of complex group interactions in collaboration with Life Campus on a Novo Nordisk Foundation-funded project.

University of Copenhagen
Research Assistant

Copenhagen, Denmark
Jun 2023–Feb 2024

- Prototyped sociometric badges and AprilTag-based spatial tracking for multimodal learning analytics.
- Built early data collection and processing workflows for mBox, a multimodal sensing prototype that later evolved into OpenMMLA.

TEACHING EXPERIENCE

University of Copenhagen

Teaching Assistant, Introduction to Python Programming and Data Science

PROFESSIONAL EMPLOYMENT

May 2019–Aug 2019 Yonyou, IT Consultant, Hong Kong, China

Jun 2017–Aug 2017 Sina.com, Data Analyst Intern, Beijing, China

PROJECTS

CoLA: A wearable multimodal AI platform for egocentric sensing, real-time behavioral analysis, and interactive facilitation in collaborative learning settings, [link](#)

OpenMMLA: An open-source IoT toolkit for multimodal data collection, synchronization, and analytics across real-world collaborative environments, [link](#)

mBox: An early multimodal sensing prototype featuring sociometric badges, audio pipelines, and AprilTag-based spatial tracking; later evolved into OpenMMLA, [link](#)

MMLA-Audio: A real-time speech analytics pipeline with speaker recognition and speech overlap detection, optimized for edge deployment, [link 1](#), [link 2](#)

MotionMatching: A real-time motion matching system on Unity implemented in C#, [link](#)

CasperFFG: A python implementation of the CasperFFG consensus combined with PoW consensus on simulated blockchain, [link](#)

SERVICE

Academic Journal Peer Review

Knowledge-Based Systems

Conference Peer Review

European Conference on Technology-Enhanced Learning (EC-TEL)

Updated September 2026